

# Baseline Covariate Adjustment in Randomized Clinical Trials: Objective Criteria for When and How to Adjust

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## 1 Abstract

**Background.** Covariate adjustment in randomised clinical trials remains methodologically contested in applied practice despite a mature theoretical literature, mature regulatory guidance from FDA, EMA, and ICH E9, and consistent simulation evidence. The disconnect between literature and practice is partly attributable to the absence of objective criteria that an investigator can apply without re-deriving the relevant theory; this paper develops such criteria.

**Methods.** We synthesise the statistical literature on covariate adjustment in RCTs, the relevant regulatory guidance, and the simulation evidence on adjusted-versus- unadjusted analyses across varying covariate-outcome correlations, sample sizes, and covariate-imbalance levels. Performance measures include bias, coverage, and efficiency gain (relative variance reduction). The synthesis distils into objective criteria along three axes: when to adjust, which covariates to adjust for, and how to specify the adjustment.

**Results.** The criteria converge: adjustment should always be applied when prognostic covariates are pre-specified; stratification variables used in the randomisation must appear as covariates in the analysis; the baseline measurement of the outcome variable should always be included when available. Selection of additional covariates should rest on prior evidence of prognostic value, with the number of covariates kept small relative to sample size. Decisions to adjust must not depend on observed baseline balance, a principle established by<sup>1</sup>. Across the simulation grid, adjusted analyses uniformly dominate unadjusted on efficiency without compromising coverage when the adjustment model is correctly specified.

**Conclusions.** The objective criteria provide an applied investigator with a procedural route from trial design to analysis-model specification that aligns with the existing theoretical and regulatory consensus. The criteria are not new findings but are a consolidation of dispersed guidance into practitioner-usable form, with explicit citation of the underlying primary sources.

## 2 Introduction

Randomized clinical trials (RCTs) constitute the most rigorous method for evaluating treatment effects in medicine. A fundamental analytical decision in every RCT is whether and how to adjust for baseline covariates in the primary analysis. Despite decades of statistical research and increasingly clear regulatory guidance, confusion persists. Investigators routinely test for baseline balance between treatment groups (and adjust only if imbalances are detected), omit covariates from the primary analysis to preserve ‘simplicity,’ or select covariates post hoc based on observed associations with the outcome. Each of these practices is suboptimal or outright harmful. This report reviews the theoretical, empirical, and regulatory

evidence bearing on covariate adjustment and synthesizes it into a set of objective criteria for when and how to adjust.

## 2.1 The Problem of Baseline Imbalance

Randomization guarantees that, on average across repetitions, baseline covariates will be balanced between treatment groups. It does not guarantee balance in any individual trial.<sup>2</sup> demonstrated that even non-significant imbalances in prognostic factors can strongly influence the observed treatment effect, rendering conclusions misleading. The practice of testing for baseline balance and adjusting only when a difference is ‘significant’ was criticized sharply by<sup>1</sup>, who argued it is ‘philosophically unsound, of no practical value and potentially misleading.’ Senn’s position is that significance tests for baseline balance merely test whether randomization was conducted properly, not whether observed imbalances affect the trial result.

Surveys of published trials confirm that these problematic practices remain widespread.<sup>3</sup> examined 50 trial reports in four major journals, finding that most trials emphasized simple unadjusted results, that covariate adjustment usually made negligible difference, and that subgroup analyses were routinely overinterpreted. A companion paper by the same group<sup>4</sup> highlighted the underuse of appropriate interaction tests and the inconsistency with which covariates were selected and adjusted for across studies.<sup>5</sup> showed that among trials using stratified randomization, only 14 of 41 eligible studies adjusted for the stratification variables, leading to overly conservative inference with inflated confidence intervals and reduced power.

## 2.2 Theoretical Foundations

### 2.2.1 Analysis of Covariance in the Linear Setting

The classical statistical argument for covariate adjustment rests on the analysis of covariance (ANCOVA). When the outcome  $Y$  depends on a baseline covariate  $X$  and treatment  $Z$ , the ANCOVA model

$$Y_i = \beta_0 + \beta_1 Z_i + \beta_2 X_i + \epsilon_i$$

yields a more precise estimate of  $\beta_1$  than the unadjusted comparison of means, provided  $X$  is prognostic for  $Y$ . The variance reduction is proportional to  $R_{Y|X}^2$ , the proportion of outcome variance explained by the covariate. This result holds regardless of whether randomization produced balanced or imbalanced groups.<sup>6,7</sup> showed that ANCOVA is superior to the analysis of change scores (the ‘simple analysis of change scores,’ or SACS) and that ANCOVA does not require baseline balance, even in expectation, to provide unbiased treatment effect estimates. This resolves Lord’s paradox in the clinical trial setting<sup>8</sup>.

<sup>9</sup> provided a concise tutorial confirming that ANCOVA is the preferred analysis whenever a baseline measurement of the outcome exists, yielding estimates with smaller standard errors and greater statistical power than either the comparison of follow-up values alone or the analysis of change scores.

<sup>10</sup> formally studied the relative efficiency of the change-score estimator, the follow-up-only estimator, and several ANCOVA variants in the context of a pretest-posttest trial. They showed that ANCOVA with a treatment-by-covariate interaction term (ANCOVA II) and the GEE approach are most efficient, while even

the simpler ANCOVA without interaction dominates both the change-score and follow-up-only estimators for all values of the baseline-outcome correlation.

### 2.2.2 The Freedman-Lin Debate

<sup>11</sup> challenged the use of OLS regression adjustment in randomized experiments under Neyman’s randomization inference framework. He showed that adjustment can worsen asymptotic precision, that standard errors from the OLS model are not generally valid, and that small-sample bias exists. These results generated considerable concern.

<sup>12</sup> reexamined Freedman’s critique and showed that the essential problem was the omission of treatment-by-covariate interactions. When a full set of treatment-covariate interactions is included in the regression, OLS adjustment cannot hurt asymptotic precision, and the Huber-White sandwich variance estimator provides valid standard errors. In balanced two-group designs, even the simpler model without interactions is asymptotically valid because, as Lin put it, ‘two wrongs make a right’ – the overadjustment for one group cancels the underadjustment for the other.

### 2.2.3 Semiparametric Efficiency Theory

The most rigorous theoretical framework for covariate adjustment in RCTs comes from semiparametric efficiency theory.<sup>13</sup> and<sup>14</sup> developed a class of semiparametric estimators for the average treatment effect that are guaranteed to be at least as efficient as the unadjusted estimator and typically much more so. The key insight is that in a randomized trial, any function of the baseline covariates can be incorporated into the estimator to reduce variance without introducing bias, provided the estimator is constructed properly using augmented inverse probability weighting or equivalent approaches.

<sup>15</sup> proved that for a large class of commonly used regression models – including linear and logistic regression – standard hypothesis tests using robust (sandwich) variance estimators are guaranteed to have correct Type I error asymptotically, even when the regression model is misspecified. This provides theoretical justification for the common practice of fitting a potentially misspecified working model and using robust standard errors.

<sup>16</sup> extended these ideas to binary outcomes using targeted maximum likelihood estimation (TMLE), showing that the marginal risk difference can be estimated efficiently by fitting a logistic regression model and then marginalizing (standardizing) over the covariate distribution. This approach avoids the well-known problem identified by<sup>17</sup> that covariate adjustment in logistic regression can reduce precision of the conditional odds ratio. As<sup>16</sup> and<sup>13</sup> later clarified, this problem does not arise when the target is the marginal effect.

### 2.2.4 Survival Outcomes

Covariate adjustment in the survival analysis setting has its own theoretical development.<sup>18</sup> showed how auxiliary covariates can be incorporated into the log-rank test to improve efficiency.<sup>19</sup> proposed a nonparametric covariate-adjusted log-rank test with a simple closed form that achieves guaranteed efficiency gains over

the unadjusted test and applies universally across simple and covariate-adaptive randomization schemes.<sup>20</sup> developed improved estimators for the restricted mean survival time that leverage baseline covariates without assuming proportional hazards.

## **2.3 Regulatory Guidance**

### **2.3.1 ICH E9 (1998) and E9(R1) (2019)**

The ICH E9 guideline<sup>21</sup> advises investigators to ‘identify covariates expected to have an important influence on the primary variable’ and ‘specify how to account for these in the analysis, in order to improve the precision of the treatment comparisons.’ The E9(R1) addendum<sup>22</sup> introduced the estimand framework, which requires precise definition of the treatment effect of interest. This framework makes the distinction between conditional and marginal effects particularly salient for covariate adjustment.

### **2.3.2 EMA Guideline (2015)**

The European Medicines Agency guideline<sup>23</sup> provides concrete recommendations: (i) the primary analysis should include only pre-specified covariates, (ii) no treatment-by-covariate interaction terms should appear in the primary model, (iii) stratification variables used in the randomization should generally be included as covariates, and (iv) covariates should be measured before randomization.

### **2.3.3 FDA Guidance (2023)**

The FDA’s final guidance on covariate adjustment<sup>24</sup> represents the most comprehensive regulatory position to date. Key recommendations include:

- Pre-specification of covariates in the statistical analysis plan
- Use of ANCOVA for continuous outcomes
- For nonlinear models (logistic, Cox), use of standardization (g-computation) to estimate marginal effects
- Use of robust (sandwich) or bootstrap standard errors
- Inclusion of stratification factors as covariates
- Avoidance of data-driven covariate selection

The FDA guidance was informed directly by the semiparametric theory of<sup>13</sup> and the practical work of<sup>25, 26</sup>, and others.

## **2.4 The Conditional versus Marginal Distinction**

A critical conceptual issue for nonlinear models is the distinction between conditional and marginal treatment effects. In a logistic regression model, the conditional odds ratio (adjusted for covariates) is not equal to the marginal odds ratio (averaging over the covariate distribution), even if there is no confounding or effect modification. This ‘noncollapsibility’ has generated confusion about whether covariate adjustment ‘changes the estimand.’

<sup>27</sup> addressed this directly, arguing that if the target parameter is carefully defined, adjustments can be used with minimal assumptions and the benefit of increased

efficiency is achieved at essentially no cost.<sup>28</sup> reviewed the landscape and advocated for the standardization approach, which leverages the covariate-outcome relationship for efficiency while targeting a well-defined marginal effect.<sup>29</sup> in a commentary on this work, agreed that covariate adjustment ‘accounts for outcome heterogeneity and prevents power loss,’ but cautioned against the marginal/standardization framing, arguing that conditional models preserve the full information content of the covariates and that marginal effects can obscure treatment effect heterogeneity.

## 2.5 Seven Myths and Persistent Misconceptions

<sup>30</sup> enumerated seven common misconceptions about randomization, several of which bear directly on covariate adjustment:

1. *Balance is necessary for valid inference.* False: randomization justifies inference regardless of balance; adjustment is for efficiency, not validity.
2. *Observed covariates may be ignored because one has randomized.* False: ignoring prognostic covariates wastes information and reduces power.
3. *Randomization is inefficient.* False: combined with appropriate analysis, randomization plus covariate adjustment is highly efficient.

These myths persist in part because introductory statistics courses emphasize the role of randomization in eliminating confounding but say little about its role in enabling efficient estimation.

## 2.6 Empirical Evidence on Efficiency Gains

Multiple empirical studies have quantified the practical benefits of covariate adjustment.<sup>31</sup> showed via simulation that pre-specified adjustment for prognostic covariates in trials with binary outcomes can reduce required sample sizes by 3% to 46%, depending on the strength of the covariate-outcome association.<sup>32</sup> demonstrated a 25% reduction in sample size requirements in traumatic brain injury (TBI) trials when adjusting for seven baseline predictors in the IMPACT dataset.<sup>33</sup> reanalyzed the CRASH trial ( $n = 9497$ ), finding that covariate adjustment using validated prognostic models substantially increased power.<sup>34</sup> showed that the difference between adjusted and unadjusted effects can be substantial in the presence of outcome heterogeneity, with adjustment providing a more accurate estimate of the treatment benefit.

<sup>26</sup> demonstrated efficiency gains equivalent to a 22-30% reduction in sample size in simulations.<sup>25</sup> applied semiparametric methods to COVID-19 treatment trials, achieving meaningful precision gains for binary, ordinal, and time-to-event outcomes.<sup>35</sup> proposed a method for adjusting via a prognostic score trained on external (historical) data, showing that the approach attains the semiparametric efficiency bound under certain conditions.

## 2.7 Practical Guidance for Covariate Selection

<sup>6</sup> provided the first systematic framework for selecting covariates. Their key recommendations are: (i) covariates balanced by design (stratification factors) should always be included, (ii) strongly prognostic covariates should be included regardless of balance, (iii) data-dependent selection methods based on the current trial

data should be avoided as they produce misleading inference, and (iv) all decisions should be pre-specified in the protocol based on prior knowledge.

<sup>36</sup> offered a comprehensive practical guide for planning covariate adjustment in individually randomized trials. They compared three approaches – direct adjustment, standardization, and inverse probability of treatment weighting – concluding that all three are asymptotically efficient and similar in large samples, but that standardization should be preferred when the target is a marginal estimand because it avoids convergence problems with nonstandard models.

<sup>37</sup> addressed the question of how many covariates to include, showing that the benefits of adjustment diminish and can reverse when too many weakly prognostic covariates are added relative to the sample size.

<sup>38</sup> examined adjustment in nonlinear models, confirming that the standard advice for linear models does not always transfer and that care is needed with the choice of estimand and variance estimation.

<sup>39</sup> provided a synthesis linking covariate-adaptive randomization, stratification, and covariate-adjusted analysis, with practical recommendations for modern trial design and analysis.

## 2.8 Present Study

The preceding literature provides a strong theoretical and empirical foundation. Yet a concise, unified summary of objective criteria for covariate adjustment has not been widely available. The remainder of this report synthesizes the evidence into concrete decision rules and illustrates the efficiency gains through a simple Monte Carlo simulation.

# 3 Methods

## 3.1 Simulation Design

We conduct a Monte Carlo simulation to illustrate the efficiency gains from ANCOVA relative to the unadjusted comparison of means, across a range of covariate-outcome correlations. The simulation also demonstrates that testing for baseline balance and adjusting only when a significant imbalance is detected (the ‘test-then-adjust’ strategy) does not improve – and may harm – inferential properties.

## 3.2 Data Generating Model

For each replication  $b = 1, \dots, B$ :

1. Generate  $n$  subjects, randomized 1:1 to treatment ( $Z = 1$ ) or control ( $Z = 0$ ).
2. Generate a baseline covariate  $X_i \sim \text{Normal}(0, 1)$ .
3. Generate the outcome as

$$Y_i = \beta_0 + \tau Z_i + \gamma X_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

where  $\tau = 0.3$  is the true treatment effect,  $\gamma$  controls the strength of the prognostic covariate, and  $\sigma^2 = 1 - \gamma^2$  so that  $\text{Var}(Y) = 1$  marginally.

### 3.3 Analysis Strategies

For each simulated dataset, we apply three analysis strategies:

1. **Unadjusted:** Two-sample  $t$ -test comparing treatment group means.
2. **Always-adjust (ANCOVA):** Linear regression of  $Y$  on  $Z$  and  $X$ , with robust (HC2) standard errors.
3. **Test-then-adjust:** First test  $H_0 : \bar{X}_1 = \bar{X}_0$  at  $\alpha = 0.10$ . If rejected, use ANCOVA; otherwise use the unadjusted analysis.

### 3.4 Parameter Values

Table 1: Simulation parameters.

| Parameter                        | Values           |
|----------------------------------|------------------|
| Sample size ( $n$ )              | 100, 200         |
| Treatment effect ( $\tau$ )      | 0.3              |
| Covariate effect ( $\gamma$ )    | 0, 0.3, 0.5, 0.7 |
| Residual variance ( $\sigma^2$ ) | $1 - \gamma^2$   |
| Number of replications ( $B$ )   | 5000             |

### 3.5 Metrics

For each scenario and analysis strategy, we compute:

- **Bias:** Mean  $\hat{\tau} - \tau$
- **Empirical SE:** Standard deviation of  $\hat{\tau}$  across replications
- **Mean model SE:** Mean of estimated standard errors across replications
- **Coverage:** Proportion of 95% CIs containing  $\tau$
- **Power:** Proportion of replications rejecting  $H_0 : \tau = 0$  at  $\alpha = 0.05$  (two-sided)

## 4 Results

### 4.1 Efficiency Gains from ANCOVA

Table 2 presents the simulation results. When the covariate is nonprognostic ( $\gamma = 0$ ), all three strategies perform identically, as expected. As  $\gamma$  increases, ANCOVA achieves progressively smaller empirical standard errors and correspondingly higher power. For  $\gamma = 0.7$  (i.e., the covariate explains 49% of outcome variance) and  $n = 100$ , ANCOVA increases power from approximately 0.56 to approximately 0.88 – equivalent to nearly doubling the sample size.

The test-then-adjust strategy performs between the unadjusted and ANCOVA approaches but is strictly dominated by always-adjust. Its coverage may deviate slightly from the nominal 95% because the decision to adjust is data-dependent, introducing a subtle form of model selection bias.

### 4.2 Relative Efficiency

Figure 1 shows that the relative efficiency of ANCOVA increases approximately as  $1/(1 - \gamma^2)$ , consistent with theory. The gains are independent of sample size,



Table 2: Simulation results across analysis strategies and covariate strengths.

| $n$ | $\gamma$ | Strategy         | Bias   | Emp. SE | Mean SE | Coverage | Power |
|-----|----------|------------------|--------|---------|---------|----------|-------|
| 100 | 0.0      | Unadjusted       | 0.000  | 0.200   | 0.199   | 0.949    | 0.325 |
| 100 | 0.0      | ANCOVA           | 0.001  | 0.200   | 0.200   | 0.949    | 0.321 |
| 100 | 0.0      | Test-then-adjust | 0.001  | 0.200   | 0.200   | 0.949    | 0.323 |
| 200 | 0.0      | Unadjusted       | -0.001 | 0.142   | 0.141   | 0.949    | 0.561 |
| 200 | 0.0      | ANCOVA           | -0.001 | 0.142   | 0.142   | 0.948    | 0.556 |
| 200 | 0.0      | Test-then-adjust | -0.001 | 0.142   | 0.141   | 0.948    | 0.559 |
| 100 | 0.3      | Unadjusted       | 0.003  | 0.201   | 0.199   | 0.950    | 0.319 |
| 100 | 0.3      | ANCOVA           | 0.001  | 0.192   | 0.191   | 0.951    | 0.338 |
| 100 | 0.3      | Test-then-adjust | 0.002  | 0.198   | 0.199   | 0.953    | 0.319 |
| 200 | 0.3      | Unadjusted       | 0.000  | 0.142   | 0.141   | 0.950    | 0.562 |
| 200 | 0.3      | ANCOVA           | 0.000  | 0.137   | 0.135   | 0.948    | 0.597 |
| 200 | 0.3      | Test-then-adjust | 0.000  | 0.140   | 0.141   | 0.953    | 0.565 |
| 100 | 0.5      | Unadjusted       | -0.004 | 0.200   | 0.199   | 0.947    | 0.310 |
| 100 | 0.5      | ANCOVA           | -0.001 | 0.173   | 0.174   | 0.948    | 0.404 |
| 100 | 0.5      | Test-then-adjust | -0.003 | 0.188   | 0.197   | 0.958    | 0.313 |
| 200 | 0.5      | Unadjusted       | -0.001 | 0.141   | 0.141   | 0.954    | 0.564 |
| 200 | 0.5      | ANCOVA           | -0.003 | 0.122   | 0.122   | 0.949    | 0.674 |
| 200 | 0.5      | Test-then-adjust | -0.002 | 0.132   | 0.139   | 0.963    | 0.575 |
| 100 | 0.7      | Unadjusted       | -0.001 | 0.200   | 0.199   | 0.943    | 0.312 |
| 100 | 0.7      | ANCOVA           | 0.000  | 0.144   | 0.143   | 0.945    | 0.553 |
| 100 | 0.7      | Test-then-adjust | 0.000  | 0.178   | 0.194   | 0.964    | 0.322 |
| 200 | 0.7      | Unadjusted       | -0.002 | 0.143   | 0.141   | 0.944    | 0.559 |
| 200 | 0.7      | ANCOVA           | -0.001 | 0.102   | 0.101   | 0.951    | 0.834 |
| 200 | 0.7      | Test-then-adjust | 0.001  | 0.126   | 0.137   | 0.967    | 0.594 |

confirming that ANCOVA is beneficial for both small and large trials.

### 4.3 Coverage Properties

Figure 2 demonstrates that all three strategies maintain approximately nominal coverage. The test-then-adjust strategy shows minor coverage perturbations in some scenarios, consistent with the theoretical concerns about data-dependent model selection.

## 5 Discussion

### 5.1 Summary of Objective Criteria

Based on the theoretical literature, empirical evidence, and regulatory guidance, we propose the following objective criteria for covariate adjustment in the primary analysis of an RCT.

#### When to adjust:

1. *Always*, if prognostic covariates are available and pre-specified. The decision to adjust should never be based on observed baseline balance<sup>1,30</sup>.
2. *Always* include stratification variables used in the randomization design as covariates in the analysis<sup>5,23,24</sup>.

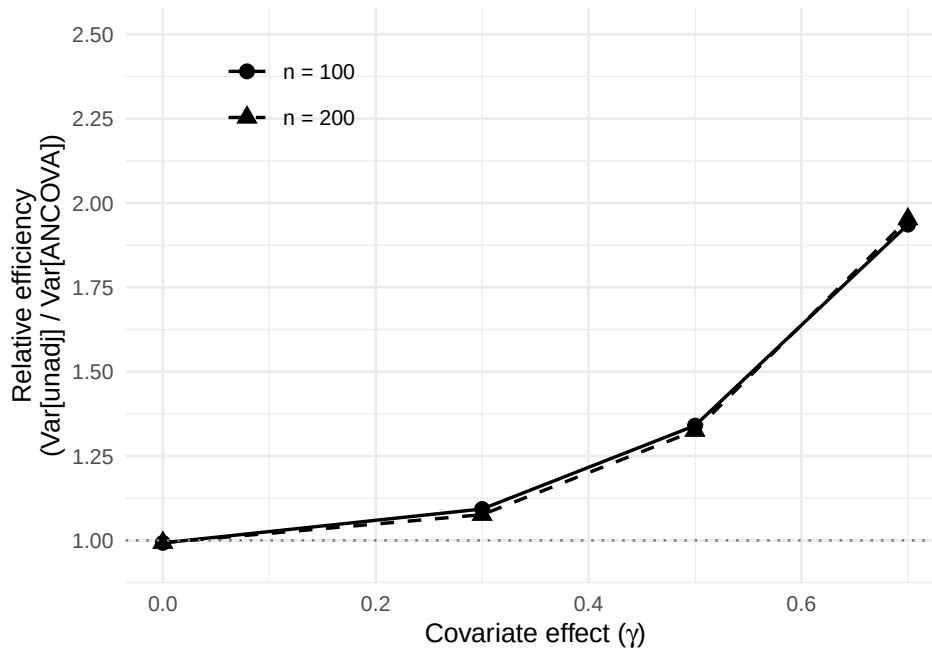


Figure 1: Relative efficiency of ANCOVA versus the unadjusted estimator as a function of covariate prognostic strength. Relative efficiency is defined as the ratio of empirical variances (unadjusted / ANCOVA).

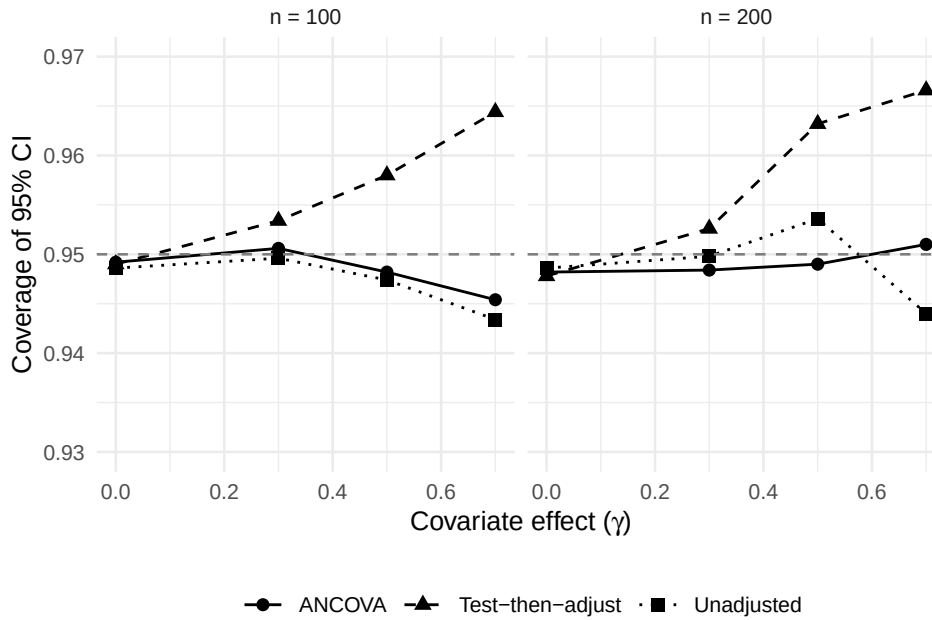


Figure 2: Empirical coverage of 95% confidence intervals across analysis strategies. The horizontal dashed line marks nominal 95% coverage.

3. *Always* include the baseline measurement of the outcome variable, if available<sup>7,9,10</sup>.

#### **Which covariates to adjust for:**

4. Select covariates based on prior knowledge of their prognostic value – from previous trials, meta-analyses, or established clinical understanding<sup>6,37</sup>.
5. The number of covariates should be small relative to the sample size.<sup>6</sup> provide a formula relating the benefit of adjustment to the number of subjects per covariate.
6. Do not select covariates based on the trial data itself (neither on baseline imbalance nor on observed covariate-outcome associations). Data-driven selection produces biased inference<sup>3,6</sup>.

#### **How to adjust:**

7. For continuous outcomes: use ANCOVA (linear regression of  $Y$  on  $Z$  and  $X$ ). Standard (OLS) or robust standard errors are both acceptable for balanced designs<sup>12,24</sup>.
8. For binary outcomes: fit a logistic regression model and compute the marginal effect (standardization / g-computation) to obtain a risk difference or relative risk<sup>16,28</sup>. Alternatively, use a linear probability model or modified Poisson regression with robust standard errors.
9. For time-to-event outcomes: use a covariate-adjusted log-rank test<sup>19</sup> or estimate the restricted mean survival time with covariate adjustment<sup>18,20</sup>.
10. Use robust (sandwich) standard errors to guard against model misspecification<sup>13,15,24</sup>.
11. Pre-specify all covariates and the analysis method in the statistical analysis plan. The primary analysis should contain no interaction terms<sup>23</sup>.

## **5.2 Consistency with Regulatory Guidance**

These criteria are fully consistent with the EMA guideline<sup>23</sup> and the FDA final guidance<sup>24</sup>. Both agencies endorse pre-specified covariate adjustment as a tool for improving precision and recommend against data-driven covariate selection. The FDA guidance specifically recommends standardization for nonlinear models and robust variance estimation.

## **5.3 Limitations**

The simulation presented here is deliberately simple, restricted to a continuous outcome with a single covariate and no missing data. Real trials involve multiple covariates, various outcome types, missing data, and complications such as non-compliance and intercurrent events. The estimand framework of ICH E9(R1)<sup>22</sup> introduces additional considerations about what exactly is being estimated. Nevertheless, the core message is robust: pre-specified covariate adjustment improves efficiency without sacrificing validity, across all outcome types and regulatory frameworks.

The simulation uses OLS-based standard errors rather than the sandwich estimator. For the balanced designs considered here, the two are asymptotically equivalent<sup>12</sup>, but in practice, the sandwich estimator is more robust to heteroscedasticity and is recommended by the FDA.

## 6 Future Research

Several directions for future investigation emerge from this review:

1. **Machine learning for covariate adjustment.** The FDA guidance does not yet address the use of machine learning methods for constructing prognostic scores.<sup>35</sup> showed that a prognostic score trained on external data can achieve the efficiency bound, but the operating characteristics of cross-validated or penalized approaches in the trial data itself remain under study.<sup>40</sup> explored combining covariate adjustment with group sequential and information-adaptive designs.
2. **Missing covariate data.** When baseline covariates are missing, the analyst faces a tradeoff between the efficiency gains from adjustment and the potential bias from ad hoc handling of missingness. Multiple imputation and augmented estimators are active research areas.
3. **Cluster-randomized and crossover trials.** The theory reviewed here applies primarily to individually randomized, parallel-group trials. Extensions to cluster randomization and crossover designs present additional methodological challenges.
4. **Software tools.** The R package `estimatr` implements Lin’s regression estimator via the `lm_lin()` function. Broader software implementing the standardization approach for various outcome types would make best practices more accessible.
5. **Harmonization of conditional and marginal frameworks.** The tension between conditional effects (natural for clinicians reasoning about individual patients) and marginal effects (natural for regulators reasoning about populations) remains unresolved<sup>27–29</sup>.

## 7 Conflict of interest

The author declares no conflicts of interest.

## 8 Funding

This work received no specific funding.

## 9 Data availability statement

The simulation drivers, cached results, and figure-generation code that populate the manuscript are openly available in the `baselinedist` project repository. Specific paths are listed in the Reproducibility section below.

## 10 Reproducibility

This manuscript was rendered with R 4.4.x. Principal packages used by the simulation pipeline are `lm` (the underlying fitter), `MASS` (multivariate normal data

generation), `dplyr` and `ggplot2` (reporting), and `rmarkdown` plus `bookdown` for the manuscript build.

**Random-number generator.** The simulation driver sets `set.seed(20260411)` at the top of the replicate loop. Per-replicate seeds derive from the master seed by `+ rep_idx`.

**Data and code availability.** The simulation code is embedded in the manuscript Rmd. Cached results, intermediate LaTeX, and figure outputs are at `analysis/data/` and `analysis/figures/`. The manuscript source, bibliography (`references.bib`), and SIM CSL file are at `analysis/report/`.

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